

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

SP6 – Financial Derivatives

May 2025

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Q.1)

- i) The payoff from this option is $\text{MAX}(S_T * X_0 - K, 0)$ where S_T is the stock price in USD on expiry X_0 is the exchange rate fixed initially and K is the strike price of the option. (3)
- ii) The underlying asset is in a foreign currency, but the option is settled in domestic currency using a fixed exchange rate. The drift of the foreign asset in domestic terms must be adjusted due to the correlation between the stock and forex rate. The objective is to find the adjusted drift r_{adj} of the foreign asset S_t under a domestic measure $\mathbb{Q}^d$ such that $V_t = S_t X_t$ grows at r_d .

Then, under $\mathbb{Q}^d$:

$$\begin{aligned}\frac{dS_t}{S_t} &= r_{\text{adj}} dt + \sigma_S d\widetilde{W}_t^S \\ \frac{dX_t}{X_t} &= (r_d - r_f) dt + \sigma_X d\widetilde{W}_t^X\end{aligned}$$

Now apply Itô to $V_t = S_t X_t$:

$$\frac{dV_t}{V_t} = (r_{\text{adj}} + (r_d - r_f) + \rho\sigma_S\sigma_X) dt + (\text{noise})$$

But V_t is in INR, so under $\mathbb{Q}^d$, it **must drift at r_d** :

$$r_d = r_{\text{adj}} + (r_d - r_f) + \rho\sigma_S\sigma_X$$

Now solve for r_{adj} :

$$r_d = r_{\text{adj}} + r_d - r_f + \rho\sigma_S\sigma_X$$

Cancel r_d from both sides:

$$0 = r_{\text{adj}} - r_f + \rho\sigma_S\sigma_X \Rightarrow \boxed{r_{\text{adj}} = r_f - \rho\sigma_S\sigma_X}$$

(7)

iii) $r_{\text{adj}} = 0.04 - 0.3 * 0.25 * 0.08 = 3.4\%$

$$d1 = \frac{\ln\left(\frac{S_t X_0}{K}\right) + (r_{\text{adj}} + 0.5 * \sigma_s^2)T}{\sigma_s \sqrt{T}} = 0.99$$

$$d2 = d1 - \sigma_s \sqrt{T} = 0.74$$

$$N(d1) = 0.84; N(d2) = 0.77$$

$$C = S_t X_0 N(d1) - K e^{-r_d T} = 2595.86$$

There can be minor rounding off errors

(5)

[15 Marks]

Q.2)

1. The price is given by:

$$Price = \frac{1}{r^{10}} \left[E^* \frac{S_1 + S_2 + \dots + S_{10}}{10} \right] = \frac{1}{10r^{10}} [E^*(S_1) + E^*(S_2) + \dots + E^*(S_{10})]$$

Now, note that $E^*[S_t] = S_0 r^t$ because in the risk-neutral probabilities, the stock price grows at the risk free rate.

Thus, the price becomes

$$Price = \frac{1}{10r^{10}} [(S_0 r) + (S_0 r^2) + \dots + (S_0 r^{10})] = \frac{S_0}{10r^{10}} [r + r^2 + \dots + r^{10}]$$

[7 Marks]

Q.3)

i) **The Vasicek model and its statistical properties:**

- This is a model used for modelling the short rate of interest $r(t)$.
- It assumes that $r(t)$ has the dynamics of an Ito process (in fact, an Ornstein-Uhlenbeck process) under the risk-neutral probability measure.
- It is a one-factor model because it incorporates a single source of randomness.
- It assumes the model $dr(t) = \alpha[\mu - r(t)]dt + \sigma dW(t)$, where $W(t)$ is standard Brownian motion
- The movements in the interest rate are therefore normally distributed (and hence can take negative values).
- The parameter σ determines the volatility.
- The parameter α is assigned a positive value, so that $r(t)$ is stationary and mean reverts to the constant value μ . A higher value of α implies more rapid mean reversion.
- The model allows negative interest rates which is generally considered undesirable.

(3)

ii)

a. We can rewrite the given function as:

$$\begin{aligned} f(t) &= r_0 e^{-\alpha t} + r_\infty (1 - e^{-\alpha t}) + k(1 - e^{-\alpha t})e^{-\alpha t} \\ &= r_\infty + (r_0 - r_\infty + k)e^{-\alpha t} - k e^{-2\alpha t} \end{aligned}$$

Differentiating,

$$\begin{aligned} f'(\tau) &= -(r_0 - r_\infty + k)\alpha e^{-\alpha t} + 2k\alpha e^{-2\alpha t} \\ &= [-(r_0 - r_\infty + k) + 2k e^{-\alpha t}]\alpha e^{-\alpha t} \end{aligned}$$

For a stationary point, this must equal zero and so the expression in square brackets must also equal zero. Thus

$$-(r_0 - r_\infty + k) + 2k e^{-\alpha t} = 0 \Rightarrow e^{-\alpha t} = \frac{r_0 - r_\infty + k}{2k}$$

b. Graph (1) will have a stationary point if

$$e^{-\alpha t} = \frac{0.04 - 0.06 + 0.1}{2(0.1)} = 0.4$$

There will be one solution to this equation, corresponding to a positive value of τ .

Also when $\tau = 0$, the slope is

$$f'(0) = [-(r_0 - r_\infty + k) + 2k]\alpha = [k - (r_0 - r_\infty)]\alpha = [0.1 - (0.04 - 0.06)]\alpha > 0$$

So graph (1) will be a humped graph starting at 0.04 (4%) for the short rate and ending at 0.06 (6%) for the long rate.

Graph (2) will have the stationary point if

$$e^{-\alpha\tau} = \frac{0.06 - 0.04 + 0.02}{2(0.02)} = 1$$

So this will have a stationary point at $\tau = 0$ only.

So Graph (2) will start at 0.06 (6%) and then fall steadily to 0.04 (4%).

- c. The usual explanation for a humped yield curve is market segmentation. Different categories of investors are attracted to bonds with different terms (short, medium or long). The bond prices, and hence the yields within each “segment” of the market, are determined by demand and supply.

In fact, graph (1) is a common shape for the yield curve.

(6)

iii) **Advantages and limitations**

- The mean-reverting property is consistent with real-world observations.
- The model is mathematically tractable, with few parameters to estimate.
- The statistical properties are based on the familiar normal distribution.
- A variety of shapes can be obtained for the yield curve, *i.e.* upward-sloping, downward sloping and humped.
- The model leads to arbitrage-free yield curves.

The model has the disadvantage that it allows $r(t)$ to take negative values, which we would not expect to observe in the real world.

(2)

[11 Marks]

Q.4) The following clues shall enable you to match the prices to the relevant contracts:

- The strike rates under (a) and (b) are very similar. But, the rate for (a) is 4% *pa* quarterly compounded corresponds to 3.969% continuously-compounded, whereas the rate for (b) is 4% *pa* monthly compounded corresponding to 3.990% *pa* continuously-compounded. Hence, the price of (b) has to be slightly lower than the price of (a).

- (d) is deeply in-the-money and thus will have the highest price
- (c) is unlike (a) and (b) but similar to (d). In fact, (c) is also in-the-money, but level of moneyness is lesser than (d). Thus, (c) is likely to have a higher price than (a) and (b) but lower than the price of (d).
- The two strike prices for a collar are usually chosen to make the price zero with the two rates usually straddling the current interest rate. Thus the value of (e) is zero.
- Finally, a fixed – for – floating swap (i.e. pay fixed and receive floating) can be replicated by 1 long cap and 1 short floor (on the same interest rate). Thus, its price has to be the difference between the prices of (a) and (c). This means that the value of the swap should be price of (a) less the price of (c) = - Rs.10,000. Thus, in this instance the floating rate payer receives Rs. 10,000 which is therefore the value of (f).

Thus, the final values and the contracts match up as follows:

Contract	Price
A	Rs. 23,000
B	Rs. 22,000
C	Rs. 28,000
D	Rs. 40,000
E	Rs. 0
F	Rs. 10,000

[10 Marks]

Q.5)

i) Types of longevity hedges are

- Longevity swap
- Longevity bonds
- Principal at risk
- Survivor cap

(4)

ii) Value of initial swap is shown as the difference between B and A (i.e. B – A), which is 43,294.77

			A	B
Year	Interest Rate	Discount Factor (Discrete basis)	Best Estimate	Regulatory basis cashflow
1	5%	0.952381	100,000	110,000
2	5%	0.907029	100,000	110,000
3	5%	0.863838	100,000	110,000
4	5%	0.822702	100,000	110,000
5	5%	0.783526	100,000	110,000
Value (Sum of PVs)			432947.7	476242.4

(6)

- iii) The value of fall in the capital requirement is the difference between C and B (i.e. C – B).

			B	C
Year	Interest Rate	Discount Factor (Discrete basis)	Regulatory basis cashflow	Value assuming increase of 50% over regulatory base
1	5%	0.952381	110,000	165000
2	5%	0.907029	110,000	165000
3	5%	0.863838	110,000	165000
4	5%	0.822702	110,000	165000
5	5%	0.783526	110,000	165000
Value (Sum of PVs)			476242.4	714363.7

The value is 238121.2.

(5)

- iv) The insurer bears that risk that the counterparty providing swap might default leading to the insurance company becoming exposed to the risk of shortfall.

Regulator could be concerned that if the counterparty has other clients in the market, a sudden increase in longevity might lead to failure of the reinsurer possibly leading to a systemic impact on the industry. The risks can be reduced by the way of collateral or retaining sufficient capital to replace the longevity swap as risk capital.

(4)

- v) Collateral value is the sum of the present values of the loss given default The LGD is the difference between the base case and the capital assuming increase of 50% over regulatory base. The same is shown in the table below

			B	C			
Year	Interest Rate	Discount Factor (Discrete basis)	Best Estimate	Value assuming increase of 50% over regulatory base	Loss Given Default	PD	PD * LGD
1	5%	0.952381	100,000	165000	65,000	2%	1,300
2	5%	0.907029	100,000	165000	65,000	4%	2,600
3	5%	0.863838	100,000	165000	65,000	6%	3,900
4	5%	0.822702	100,000	165000	65,000	8%	5,200
5	5%	0.783526	100,000	165000	65,000	10%	6,500
Value (Sum of PVs)							16336.31

The collateral = 16,336.31. In real markets the collateral is usually adjusted dynamically.

(6)

[25 Marks]

Q.6)

- i) Rearrange the AR(1) process $X_t - b = a(X_{t-1} - b) + \varepsilon_t$ to yield $X_t - X_{t-1} = (1 - a)(b - X_{t-1}) + \varepsilon_t$, which is a discrete time analog of the Vasicek model. Further, the autocorrelation of X_t at lag one is a , which translates to $(1 - \beta dt)$ in the Vasicek model. (2)
- ii) Changes in slope are due to changes in a , changes in curvature are due to changes in b and parallel moves are due to changes in r . (3)
- iii) The normal model does, for the following reason: It has fatter left tails and thinner right tails for probability distributions of interest rates in the future. It therefore gives thinner left tails and fatter right tails for the probability distribution of bond prices. (2)
- iv) If every node contains forward rates up to the maturity of the underlying bond, a term structure up to that maturity can be constructed. The tree then needs to extend to only the maturity of the claim. A full tree is needed when there is long term path dependency. (2)

[9 Marks]**Q.7)**

- i) Let's evaluate both the transactions.

Selling the stock

- Receives $\$(650 * (1 - 1\%)) = \643.50
- Loss of dividend of \$45 as on May 15, 2025 whose value as on May 1, 2025 is
 - \$44.90 (if computed 5% for 15 days)
 - \$44.89 (if computed @ 6% for 15 days)

Since the client will be a net lender rather than borrower the applicable rate of interest is 5%. And hence the loss of dividend in present value terms should be \$ 44.90

Buying the futures

- Pay the brokerage: $\$(594 * 0.005) = \2.97
- Pay the margin: $\$(651 * 0.05) = \32.55
- Pay $\$(594 - 32.55) = \561.45 on maturity (May 31, 2025) and get the shares.
- The value of this amount as on May 1, 2025 is the present value of \$561.45 @ 5% for 31 days = \$ 559.08

In a nutshell, the expected gain from the strategy today is:

$$\$(643.50 - \$44.90 - \$2.97 - \$32.55 - \$559.08) = \$4.00$$

Hence, the client can be advised to go ahead with the strategy

(7)

ii) The risks in implementing this strategy are:

- The discount is not as substantial as you may infer by looking at the prices.
- Even if the prices are known for calculating the gain/loss before putting the order on the exchange, the actual transacted price may be different compared to the price used in calculating the gains.
- Taxes have been ignored in the above calculations and they may decrease/increase the benefits.
- The daily settlements have been ignored in the above calculations. This may give rise to further cash-flow management problems and interest gain/losses on the same.
- Liquidity risks may be there when markets turn volatile.

(2)

[9 Marks]

Q.8)

i)

Interest rate 12%

Initial Default Rate 4%

increase 0.50%

Recovery 40.00%

Year		Default per 100	Survival		Present value
1	4%	4	96	0.88692	0.851444
2	4.50%	4.32	91.68	0.786628	0.72118
3	0.00%	0	91.68	0.697676	0.63963
4	5.50%	5.0424	86.64	0.618783	0.536099
					2.7484

Periodic payment 2.7484S

Principal payment default take place mid of year

Year	Default per 100	Recovery	Pay off	Discount	Present value
1	4	40.00%	2.4	0.941765	0.022602
2	4.32	40.00%	2.592	0.83527	0.02165
3	0	40.00%	0	0.740818	0
4	5.0424	40.00%	3.02544	0.657047	0.019879
					0.0641

Total principal payment 0.0641

Accrual adjustment per 1 unit of spread

Year	Survival	Accrual	Discount	
1	96	2	0.941765	0.0188
2	91.68	2.16	0.83527	0.0180
3	91.68	0	0.740818	-
4	86.64	2.5212	0.657047	0.0166
			Sum	0.0534

Accrual payment			0.0534S
Spread	0.022878	$0.0641/(2.7484+0.0534)$	
	228.7815		

(8)

ii)

Interest rate	12%				
Start int	4%				
increase	0.50%				
Recovery	40.00%				
Year		Default per 100	Survival		Present value
1	4%	4	96	0.88692	0.8514
2	4.50%	4.32	91.68	0.786628	0.7212
3	5.00%	4.584	87.1	0.697676	0.6076
4	5.50%	4.79028	82.31	0.618783	0.5093
					2.6896
Periodic payment			2.6896S		

Periodic payment			2.6896S		
Principal payment	default take place mid of year				
Year	Default per 100	Recovery	Pay off	Discount	Present value
1	4	40.00%	2.4	0.941765	0.0226
2	4.32	40.00%	2.592	0.83527	0.0217
3	4.584	40.00%	2.7504	0.740818	0.0204
4	4.79028	40.00%	2.874168	0.657047	0.0189
					0.0835
Total principal payment			0.0835		

Accrual adjustment per 1 unit of spread				
Year	Survival	Accrual	Discount	
1	96	2	0.941765	0.018835
2	91.68	2.16	0.83527	0.018042
3	87.1	2.292	0.740818	0.01698
4	82.31	2.39514	0.657047	0.015737
			Sum	0.0696

Accrual payment		0.0696S
Spread	0.030262	$= (0.0835/(0.0693+2.6896))$

	302.6239		
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(6)

[14 Marks]
